



Part 3: Practicing Development of Full-Stack Web Applications

In this part, we are going to dive deeper into full-stack web development. We will start by adding authentication to our app using **JSON Web Tokens**. Then, we will learn how to improve the load time using server-side rendering. Next, we will learn how to optimize an app for search engines. We will also implement end-to-end tests using **Playwright** to make sure our app stays robust. Then, we will learn how to collect events, aggregate data using **MongoDB**, and create statistics to visualize the aggregated data using **Victory**. Toward the end of this part, we will learn about building a backend with a **GraphQL API** and how to interface with GraphQL on the frontend using **Apollo Client**.

This part includes the following chapters:

- [Chapter 6](#), *Adding Authentication with JWT*
- [Chapter 7](#), *Improving the Load Time Using Server-Side Rendering*
- [Chapter 8](#), *Making Sure Customers Find You with Search Engine Optimization*
- [Chapter 9](#), *Implementing End-To-End Tests Using Playwright*
- [Chapter 10](#), *Aggregating and Visualizing Statistics Using MongoDB and Victory*
- [Chapter 11](#), *Building a Backend with a GraphQL API*
- [Chapter 12](#), *Interfacing with GraphQL on the Frontend Using Apollo Client*

